FeignClient - Synchronous Communication

FeignClient:

- Feign is a Declarative HTTP client developed by Netflix.
- Declarative means "we tell What to do, not How to do".

In SpringBoot , Feign capability is available via Spring Cloud OpenFeign library.

- Spring Cloud provides a set of tool and libraries, which helps to build distributed microservices.
- As it provides seamless integration with :
  - Service Discovery
  - Client side Load Balancing
  - Circuit Breaker and Resilience
  - Api Gateway
  - Distributed Tracing
  - Centralized Configuration etc....

Pom.xml

Concept && Coding

```
<dependency>
    <groupId>org.springframework.cloud</groupId>
    <artifactId>spring-cloud-starter-openfeign</artifactId>
</dependency>

```

In future, we might use more Spring Cloud libraries (for Load balancer, for Service Discovery etc.) and all those Spring Cloud libraries should have compatible version, therefore we use below dependency management, so that we don't have to mange it manually.

That's why we are not specifying the version with above **"spring-cloud-starter-openfeign**" dependency, it will be taken care by our dependency management.

```
<dependencyManagement>
    <dependencies>
        <dependency>
            <groupId>org.springframework.cloud</groupId>
            <artifactId>spring-cloud-dependencies</artifactId>
            <version>2023.0.1</version> <!-- Use latest compatible version-->
            <type>pom</type>
            <scope>import</scope>
        </dependency>
    </dependencies>
</dependencyManagement>

```

OrderService ----------⟶ProductService

```
@FeignClient(
```

```
        name = "product-service",
        url = "${feign.client.product-service.url}"
)
public interface ProductClient {

    @GetMapping("/products/{id}")
    String getProductById(@PathVariable("id") String id);
}

```

```
@RestController
@RequestMapping("/products")
public class ProductController {

    @GetMapping("/{id}")
    public String getProduct(@PathVariable String id) {
        return "Product fetched with id: " + id;
    }
}

```

application.properties

```
server.port=8081

# Base URL for Product Service
feign.client.product-service.url=http://localhost:8082

```

```
@RestController
@RequestMapping("/orders")
public class OrderController {

    @Autowired
    ProductClient productClient;

    @GetMapping("/{id}")
    public ResponseEntity<String> getOrder(@PathVariable String id) {

        String responseFromProductAPI = productClient.getProductById(id);
        System.out.println(
                "Response from Product api call is: " + responseFromProductAPI
        );

        return ResponseEntity.ok("order call successful");
    }
}

```

Concept && Coding

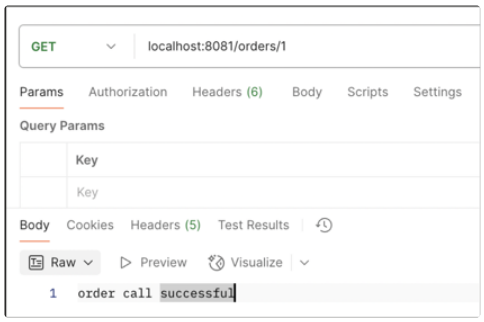
```
@SpringBootApplication
@EnableFeignClients
public class OrderserviceApplication {

    public static void main(String[] args) {
        SpringApplication.run(OrderserviceApplication.class, args);
    }
}

```

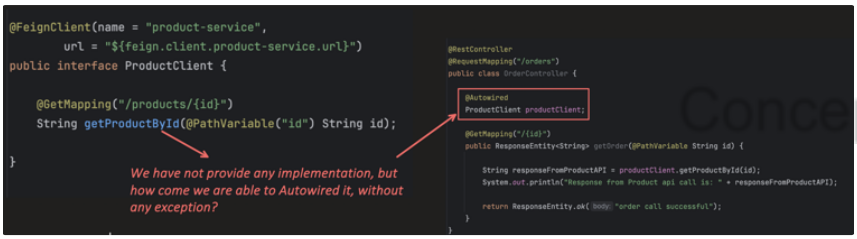
@EnableFeignClients *enable Feign support and tells SpringBoot to scan for interfaces annotated with @FeignClient*

Start the application and invoke the Order Endpoint

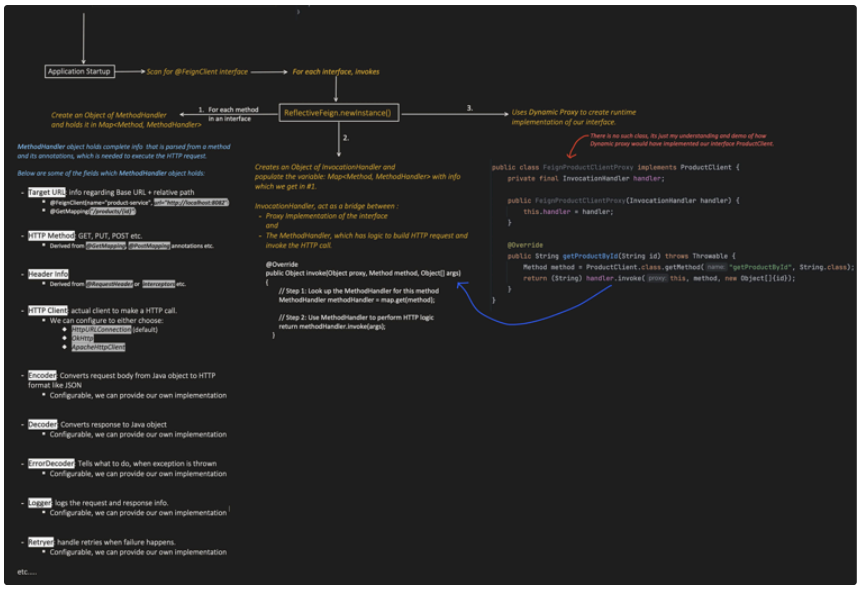



```
2025-06-06T21:11:17.438+05:30  INFO 18774 --- [          main] c.c.o.OrderserviceApplication           : Starting OrderserviceApplication using
2025-06-06T21:11:17.439+05:30  INFO 18774 --- [          main] c.c.o.OrderserviceApplication           : No active profile set, falling back to
2025-06-06T21:11:17.705+05:30  INFO 18774 --- [          main] o.s.cloud.context.scope.GenericScope    : BeanFactory id=f55ac312-243e-3904-baf4-
2025-06-06T21:11:17.810+05:30  INFO 18774 --- [          main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 8081 (http
2025-06-06T21:11:17.814+05:30  INFO 18774 --- [          main] o.apache.catalina.core.StandardService  : Starting service [Tomcat]
2025-06-06T21:11:17.814+05:30  INFO 18774 --- [          main] o.apache.catalina.core.StandardEngine   : Starting Servlet engine: [Apache Tomcat
2025-06-06T21:11:17.836+05:30  INFO 18774 --- [          main] o.a.c.c.C.[Tomcat].[localhost].[/]      : Initializing Spring embedded WebApplica
2025-06-06T21:11:17.837+05:30  INFO 18774 --- [          main] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initializat
2025-06-06T21:11:17.999+05:30  INFO 18774 --- [          main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8081 (http) with
2025-06-06T21:11:18.005+05:30  INFO 18774 --- [          main] c.c.o.OrderserviceApplication           : Started OrderserviceApplication in 0.71
2025-06-06T21:11:28.409+05:30  INFO 18774 --- [nio-8081-exec-2] o.a.c.c.C.[Tomcat].[localhost].[/]      : Initializing Spring DispatcherServlet '
2025-06-06T21:11:28.409+05:30  INFO 18774 --- [nio-8081-exec-2] o.s.web.servlet.DispatcherServlet       : Initializing Servlet 'dispatcherServlet
2025-06-06T21:11:28.410+05:30  INFO 18774 --- [nio-8081-exec-2] o.s.web.servlet.DispatcherServlet       : Completed initialization in 1 ms
Response from Product api call is: fetch the product details with id:1
```

<u>So, first important thing to understand is, how this Declarative HTTP Calls works?</u>



Concept && Coding



MethodHandler *also has method invoke(), which knows how to create HTTP request object from above info and make a HTTP call.*

```java
@Override
public Object invoke(Object[] argv) throws Throwable {
    RequestTemplate template = buildTemplateFromArgs.create(argv);
    Options options = findOptions(argv);
    Retryer retryer = this.retryer.clone();
    while (true) {
        try {
            return executeAndDecode(template, options);
        } catch (RetryableException e) {
            try {
                retryer.continueOrPropagate(e);
            } catch (RetryableException th) {
                Throwable cause = th.getCause();
                if (propagationPolicy == UNWRAP && cause != null) {
                    throw cause;
                } else {
                    throw th;
                }
            }
            if (logLevel != Logger.Level.NONE) {
                logger.logRetry(metadata.configKey(), logLevel);
            }
            continue;
        }
    }
}
```

An example below, with various annotation usage demo like @GetMapping, @PutMapping, @PathVariable, @RequetParam, @RequestHeader, @RequestBody etc..

Concept && Coding

Order Application

```java
@FeignClient(
        name = "product-service",
        url = "${feign.client.product-service.url}"
)
public interface ProductClient {

    @GetMapping("/products/{id}")
    String getProductById(@PathVariable("id") String id);

    @PutMapping(
            value = "/products/update/{id}",
            consumes = "application/json"
    )
    Product updateProduct(
      /*
          Ordering is not mandatory,
          Springboot will take care to pass
          values according to annotations.
      */
            @PathVariable("id") String id,
            @RequestParam("sendMail") boolean sendMail,
            @RequestHeader("X-ConceptCoding-ID") String uniqueID,
            @RequestBody Product updatedProductDetails
    );
}

```

Product Application

```java
@RestController
@RequestMapping("/products")
public class ProductController {

    @GetMapping("/{id}")
    public ResponseEntity<String> getProduct(@PathVariable String id) {
        return ResponseEntity.ok("fetch the product details with id:" + id);
    }

    @PutMapping("/update/{id}")
    public ResponseEntity<Product> createProduct(
            @PathVariable String id,
            @RequestBody Product product,
            @RequestParam("sendMail") boolean sendMail,
            @RequestHeader("X-ConceptCoding-ID") String uniqueID
    ) {

        Product dbProductObject = getProductFromDB(id);
        dbProductObject.setName(product.getName());

        return ResponseEntity.ok(dbProductObject);
    }
}

```

Encoder and Decoder in FeignClient

- **Encoder**: Converts a Java object into a request body (say JSON).
- **Decoder**: Converts the HTTP response body (say JSON) into a Java object.

Concept && Coding

If we want our custom Encoder and Decoder implementation:

```java
@FeignClient(
        name = "product-service",
        url = "${feign.client.product-service.url}",
        configuration = ProductClientConfig.class
)
public interface ProductClient {

    @PutMapping(
            value = "/products/update/{id}",
            consumes = "application/json"
    )
    Product updateProduct(
            @PathVariable("id") String id,
            @RequestParam("sendMail") boolean sendMail,
```

```
            @RequestHeader("X-ConceptCoding-ID") String uniqueID,
            @RequestBody Product updatedProductDetails
    );
}

```

```
public class MyCustomProductClientEncoder implements Encoder {

    @Override
    public void encode(
            Object object,
            Type bodyType,
            RequestTemplate template
    ) throws EncodeException {

        try {
            String jsonString =
                    new ObjectMapper().writeValueAsString(object);
            template.body(jsonString);
        } catch (Exception e) {
            throw new EncodeException("Unable to encode object");
        }
    }
}

```

```
public class MyCustomProductClientDecoder implements Decoder {

    @Override
    public Object decode(Response response, Type type)
            throws IOException, DecodeException, FeignException {

        InputStream responseBody =
                response.body().asInputStream();

        return new ObjectMapper().readValue(
                responseBody,
                new TypeReference<Object>() {
                    @Override
                    public Type getType() {
                        return type;
                    }
                }
        );
    }
}

```

Concept && Coding

```
@Configuration
public class ProductClientConfig {

    @Bean
    public Encoder myCustomEncoder() {
        return new MyCustomProductClientEncoder();
    }

    @Bean
    public Decoder myCustomDecoder() {
        return new MyCustomProductClientDecoder();
    }
}

```

ErrorDecoder in FeignClient

- It is used to handle non 2xx status codes like 4xx and 5xx.

If we want our custom ErrorDecoder implementation:

```
@FeignClient(
        name = "product-service",
        url = "${feign.client.product-service.url}",
        configuration = ProductClientConfig.class
)
public interface ProductClient {

    @PutMapping(
            value = "/products/update/{id}",
            consumes = "application/json"
    )
    Product updateProduct(
            @PathVariable("id") String id,
            @RequestParam("sendMail") boolean sendMail,
            @RequestHeader("X-ConceptCoding-ID") String uniqueID,
            @RequestBody Product updatedProductDetails
    );
}

```

Concept && Coding

```
public class MyCustomProductClientErrorDecoder implements ErrorDecoder {

    private final ErrorDecoder defaultErrorDecoder = new Default();

    @Override
    public Exception decode(String methodKey, Response response) {

        HttpStatus statusCode = HttpStatus.valueOf(response.status());

        if (statusCode.is4xxClientError()) {
            return new MyCustomBadRequestException("Client Error");
        } else if (statusCode.is5xxServerError()) {
            return new MyCustomServerException("Server Error");
        } else {
            return defaultErrorDecoder.decode(methodKey, response);
        }
    }
}

```

```
@Configuration
public class ProductClientConfig {

    @Bean
    public ErrorDecoder myCustomErrorDecoder() {
        return new MyCustomProductClientErrorDecoder();
```

```
    }
}

```

```
2025-06-07T20:26:24.770+05:30  INFO 24295 --- [nio-8081-exec-1] o.s.web.servlet.DispatcherServlet        : Completed initialization in 1 ms
2025-06-07T20:26:24.812+05:30 ERROR 24295 --- [nio-8081-exec-1] o.a.c.c.C.[.[.[/].[dispatcherServlet]     : Servlet.service() for servlet [dispatcherServlet]

com.conceptandcoding.orderservice.BadRequestException Create breakpoint : Client Error
    at com.conceptandcoding.orderservice.MyCustomProductClientErrorDecoder.decode(MyCustomProductClientErrorDecoder.java:17) ~[classes/:na]
    at feign.InvocationContext.decodeError(InvocationContext.java:126) ~[feign-core-13.2.1.jar:na]
    at feign.InvocationContext.proceed(InvocationContext.java:72) ~[feign-core-13.2.1.jar:na]
    at feign.ResponseHandler.handleResponse(ResponseHandler.java:63) ~[feign-core-13.2.1.jar:na]
```

Retryer in FeignClient

```
@Override
public Object invoke(Object[] argv) throws Throwable {
    RequestTemplate template = buildTemplateFromArgs.create(argv);
    Options options = findOptions(argv);
    Retryer retryer = this.retryer.clone();
    while (true) {
        try {
            return executeAndDecode(template, options);
        } catch (RetryableException e) {
            try {
                retryer.continueOrPropagate(e);
            } catch (RetryableException th) {
                Throwable cause = th.getCause();
                if (propagationPolicy == UNWRAP && cause != null) {
                    throw cause;
                } else {
                    throw th;
                }
            }
            if (logLevel != Logger.Level.NONE) {
                logger.logRetry(metadata.configKey(), logLevel);
            }
            continue;
        }
    }
}
```

During execute, if any exception happens, first its checked, if it can be retried or not.

Concept && Coding

- Retry only happens when there is either:
  - Connection time out
  - Network related exception like (IOException)
- After all retry finished, then ErrorDecoder is invoked.
- For 4xx and 5xx, retry do not happens, its handled by ErrorDecoder directly.

```
public interface Retryer extends Cloneable {

    if retry is permitted, return (possibly after sleeping). Otherwise, propagate the exception.
    void continueOrPropagate(RetryableException e);

    Retryer clone();

    class Default implements Retryer {

        private final int maxAttempts;
        private final long period;
        private final long maxPeriod;
        int attempt;
        long sleptForMillis;

        public Default() {      this( period: 100, SECONDS.toMillis( duration: 1),  maxAttempts: 5); }
```

- Default it uses Retryer.Default :
  - 5 times call attempt (includes the 1st call too)
  - Initial wait time between retries is 100ms
  - Wait time double with each retry
  - But max wait time can be 1second

- Try 1 : (immediate attempt)
- Try 2 : wait 100ms
- Try 3 : wait 200ms
- Try 4 : wait 400ms
- Try 5 : wait 800ms (but max capped at 1 second)

After all retry attempt finished, ErrorDecoder is invoked.

If, we don't want to retry at all, we can use Retryer.NEVER_RETRY (this already present in Retryer class)

```
@Configuration
public class ProductClientConfig {

    @Bean
    public Retryer myCustomRetryer() {
        return Retryer.NEVER_RETRY;
    }
}

```

If, we want custom implementation

```
@FeignClient(
        name = "product-service",
        url = "${feign.client.product-service.url}",
        configuration = ProductClientConfig.class
)
public interface ProductClient {

    @PutMapping(
            value = "/products/update/{id}",
            consumes = "application/json"
    )
    Product updateProduct(
            @PathVariable("id") String id,
            @RequestParam("sendMail") boolean sendMail,
            @RequestHeader("X-ConceptCoding-ID") String uniqueID,
            @RequestBody Product updatedProductDetails
    );
}

```

Concept && Coding

UserCase-1 : I only want to control Attempt, wait time and max period, rest I want to reuse the "Retryer.Default" logic.

```
public class MyCustomRetryer extends Retryer.Default {

    // control attempts & wait time, rest uses Default implementation
    public MyCustomRetryer() {
        super(200, 1000, 4); // period, maxPeriod, maxAttempts
    }
}

```

```
@Configuration
public class ProductClientConfig {

    @Bean
    public Retryer myCustomRetryer() {
        return new MyCustomRetryer();
    }
}

```

UserCase-2 : I want full control, then I have to implement Retryer itself and provide the custom implementation for the "continueOrPropogate()" method.

```java
public class MyCustomRetryer implements Retryer {

    private int attempt = 1;
    private final int maxAttempts = 5;

    @Override
    public void continueOrPropagate(RetryableException e) {
        if (attempt >= maxAttempts) {
            throw e;
        }
        attempt++;
        try {
            Thread.sleep(100);
        } catch (InterruptedException ie) {
            // ignore
        }
    }

    @Override
    public Retryer clone() {
        return new MyCustomRetryer();
    }
}

```

Last but not the least:

During start, we discussed that, this name is just a arbitrary value. And its just we are giving the name to our FeignClient.

Concept && Coding

```java
@FeignClient(
        name = "product-service",
        url = "${feign.client.product-service.url}"
)
public interface ProductClient {

    @GetMapping("/products/{id}")
    String getProductById(@PathVariable("id") String id);
}

```

But where its exactly used?

Yes, this name comes handy, when we have to provide any configuration in application.properties

If we want to set request and connection timeout only for product-service FeignClient

application.properties

```properties
# timeout applicable only to product-service FeignClient
feign.client.config.product-service.connectTimeout=3000
feign.client.config.product-service.readTimeout=5000

```

If we want to set request and connection timeout for all FeignClient

application.properties

```
# timeout applicable to all FeignClients
feign.client.config.default.connectTimeout=3000
feign.client.config.default.readTimeout=5000

```

Concept && Coding